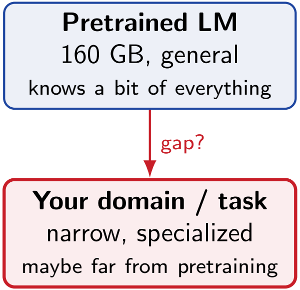
A pretrained LM is general. Your data is not.

RoBERTa was pretrained on **160 GB** of broad English text – encyclopedias, news, books, web. Strong across many tasks, so we treat it as a finished foundation.

But your target is a *slice* of language: chemical-protein abstracts, ACL citations, Amazon reviews. That slice may be **far** from anything RoBERTa saw much of.

You have a pile of *unlabeled* in-domain text and the usual fine-tuning recipe. *Before* you fine-tune – should you keep **pretraining** on that pile first?

When is a second phase of pretraining worth it, and how little of it do you need?



Outline

Setup: one general model, many specific worlds

DAPT: keep pretraining on the domain

Relevance matters: the wrong domain can hurt

TAPT: keep pretraining on the task's own text

Augmenting TAPT, and cross-task limits

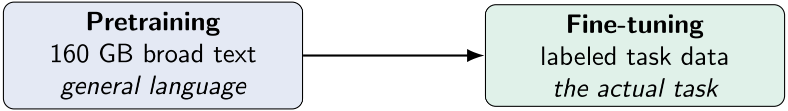
Recap

One general model, many specific worlds

RoBERTa, four domains, eight tasks – and how far each domain sits from pretraining.

The question: is “domain” still relevant?

Modern NLP is two stages: **pretrain** a big LM on massive unlabeled text, then **fine-tune** it on a labeled task.



With models this broad, is a task’s *textual domain* still worth caring about? Or do big pretrained LMs already cover everything?

> The paper inserts a **second phase of pretraining** – on unlabeled in-domain or in-task text – *between* the two stages, and measures what it buys.

The experimental grid: 4 domains $\times$ 2 tasks

Domain	Tasks (8 total)
BioMed	CHEMPROT, RCT
CS	ACL-ARC, SCIERC
News	HYPERPARTISAN, AGNEWS
Reviews	HELPFULNESS, IMDB

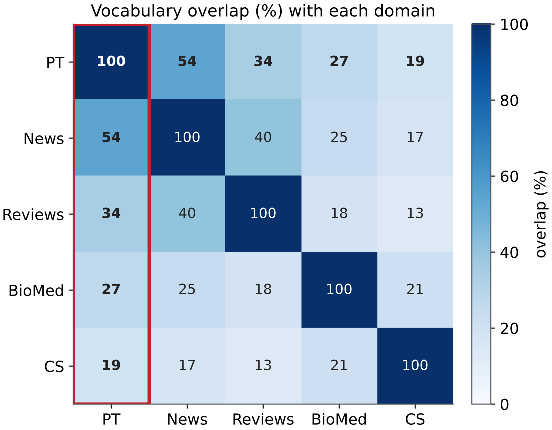
Classification tasks spanning **high-** and **low-resource** settings (some under 2K labeled examples). One shared baseline: off-the-shelf **RoBERTa-base**, fine-tuned per task.

Each domain also comes with a **large unlabeled corpus** for continued pretraining:

- BioMed: 7.6B tokens (47 GB)
- CS: 8.1B tokens (48 GB)
- News: 6.7B tokens (39 GB)
- Reviews: 2.1B tokens (11 GB)

These feed the "don't stop pretraining" phase we are about to test.

How far is each domain from pretraining?



far from PT -> more to gain from DAPT

Overlap of the top-10K words with a sample of RoBERTa's pretraining corpus (**PT**):

- News **54%**, Reviews **35%** – close to PT.
- BioMed **27%**, CS **19%** – far.

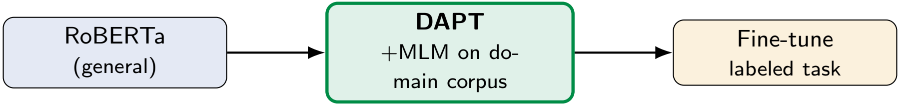
A cheap predictor: **the more dissimilar the domain, the more room a second pretraining phase has to help.**

DAPT – Domain-Adaptive PreTraining

Continue masked-LM pretraining on a big in-domain corpus, *then* fine-tune.

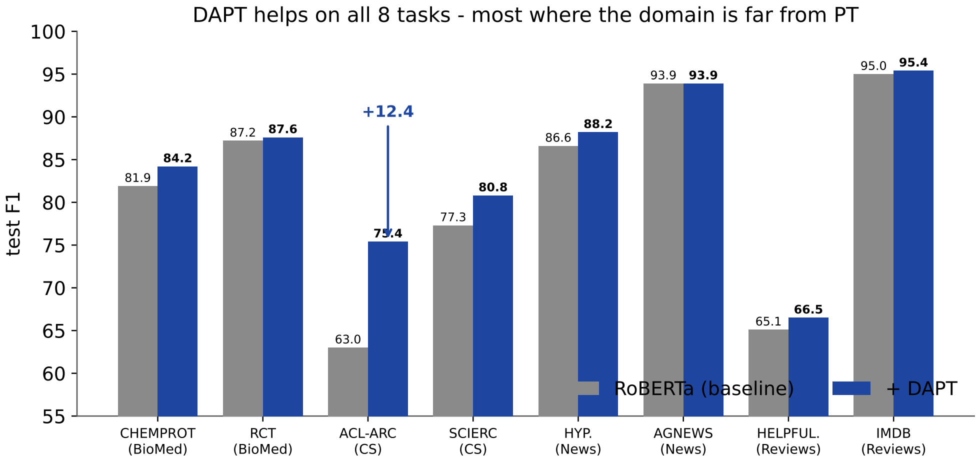
What DAPT is

Domain-Adaptive PreTraining: take off-the-shelf RoBERTa and *continue* its masked-language-modeling training on a large **unlabeled** domain corpus – one pass, 12.5K steps – *before* the usual supervised fine-tuning.



Evidence it fits the domain: masked-LM loss drops after DAPT – BioMed $1.32 \rightarrow 0.99$, CS $1.63 \rightarrow 1.34$, Reviews $2.10 \rightarrow 1.93$. (News barely moves, $1.08 \rightarrow 1.16$ – it was already in-domain for RoBERTa.)

DAPT consistently helps – most where the domain is far



DAPT beats or ties the baseline on **all 8** tasks. The gains are biggest in the **far** domains (CS, BioMed) – e.g. ACL-ARC **+12.4** F1 – and small where the domain already overlaps PT (News: AGNEWS unchanged).

Is it the domain, or just more data?

A control that separates “relevant” from “simply more text.”

Predict first: does any extra pretraining help?

DAPT works. But *why*? Maybe RoBERTa just benefits from seeing **more text**, regardless of what it is.

Control (the paper calls it $\neg$DAPT): adapt each task to an **irrelevant** domain instead – News$\rightarrow$adapt on CS, Reviews$\rightarrow$adapt on BioMed, etc. Same amount of extra pretraining, wrong content.

Prediction: does adapting to an irrelevant domain still help, do nothing, or ***hurt****?*

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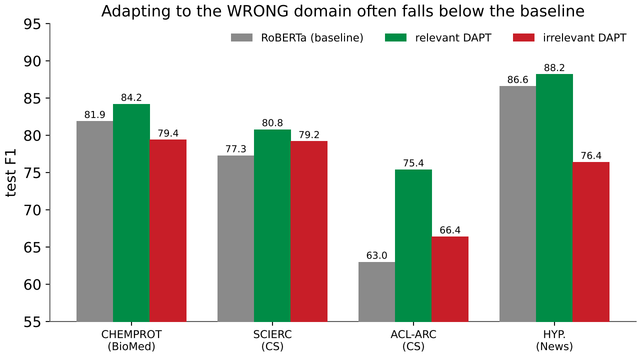
Prediction: does adapting to an irrelevant domain still help, do nothing, or ***hurt****?*

It often **hurts** – irrelevant adaptation drops *below* the plain baseline. More data is not the mechanism; **domain relevance** is.

The wrong domain drags you below baseline



Relevant DAPT (green) beats the baseline; the **irrelevant** domain (red) usually lands *under* it.

The extreme case: HYPERPARTISAN drops **86.6** $\rightarrow$ **76.4** when adapted on the wrong (CS) domain.

Exposure to *more* data without domain relevance is, in most settings, **detrimental**.

Caveat: domain boundaries are fuzzy

Our labels ("News," "Reviews") are convenient, not sharp. Figure 2 already showed **40%** vocabulary overlap between Reviews and News.

The paper finds actual cross-domain look-alikes: an IMDB review of a 1940 film and a REALNEWS article discussing the same film share vocabulary and phrasing.

Consequence: adapting Reviews tasks on *News* is **not** very harmful (it is a near neighbor), whereas adapting them on BioMed is.

"Relevant" is a **spectrum**, not a binary. The vocabulary-overlap map is a cheap first guess at where a domain sits.

TAPT – Task-Adaptive PreTraining

Forget the huge corpus – just keep pretraining on the task's *own* unlabeled text.

What TAPT is

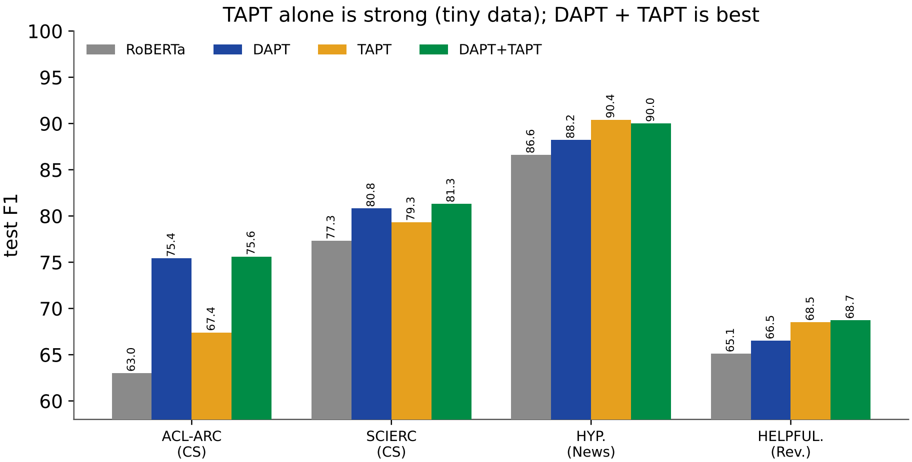
A task's dataset is a *narrow slice* of its domain (CHEMPROT is a few hand-picked PubMed categories, not all of biomedicine). So pretrain on **exactly that slice**.

Task-Adaptive PreTraining: continue masked-LM pretraining on the task's **own unlabeled training text** (the same sentences, labels removed), for 100 epochs with fresh random masks each pass – then fine-tune.

A **different trade-off** from DAPT: a *far smaller* corpus, but one that is *maximally* task-relevant.

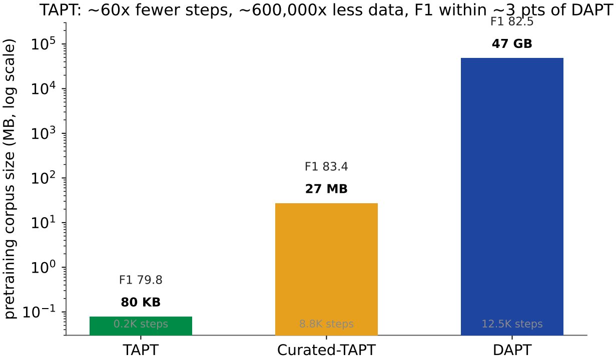
Cheap enough to run per-task – no giant domain corpus, no extra labeling. It reuses text you already have.

TAPT is strong; DAPT + TAPT is best



TAPT (orange) beats the baseline on every task and sometimes *matches or beats* DAPT (e.g. HYPERPARTISAN, HELPFULNESS) using a fraction of the data. Doing **DAPT then TAPT** (green) gets the best of both – the top score on all 8 tasks in the paper.

How TAPT is so cheap



On RCT-500, TAPT pretrains on **500 documents** (∼80 KB) in 0.2K steps – DAPT uses **25M documents** (47 GB) in 12.5K steps.

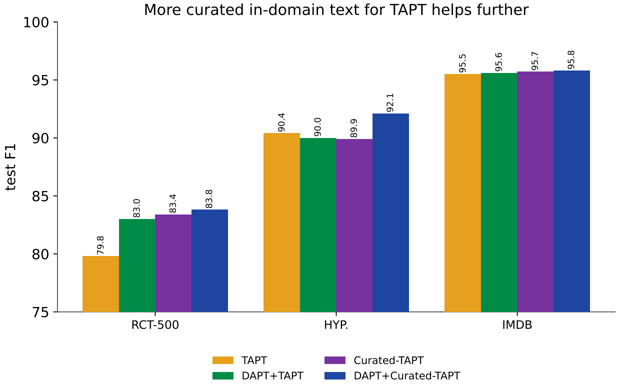
That is roughly **60**× **fewer steps** and *orders of magnitude* less storage, for an F1 within about **3 points** of DAPT.

Reuse: adapt the domain *once* (DAPT), then add a tiny TAPT phase per task.

More task-relevant text, and where TAPT stops

Feeding TAPT extra in-domain text – and why it does not transfer across tasks.

Augmenting TAPT with more in-domain text



TAPT's corpus is small – so give it **more** text from the task distribution:

- **Curated:** a large human-collected pool of unlabeled task data (Curated-TAPT).
- **Retrieved:** pull nearest-neighbor sentences from the domain corpus (kNN-TAPT) when no curated pool exists.

Both lift TAPT further; **DAPT + Curated-TAPT** is strongest.

TAPT is task-specific – it does not transfer

Because TAPT fits the *task's* distribution so tightly, a model TAPT-ed on one task and fine-tuned on *another* task in the same domain does **worse** (Transfer-TAPT):

Task	TAPT	Transfer-TAPT	Δ
HYPERPARTISAN	89.9	82.2	-7.7
HELPFULNESS	68.5	65.0	-3.5
ACL-ARC	67.4	64.1	-3.3
CHEMPROT	82.6	80.4	-2.2

Tasks within one domain have *different* distributions. This is exactly why a broad domain alone is not enough – and why **DAPT then TAPT** is complementary, not redundant.

Recap

- **Don't stop pretraining:** a second phase of masked-LM pretraining, on unlabeled text, *before* fine-tuning – reliably helps.
- **DAPT** (big in-domain corpus) helps on all 8 tasks, **most when the domain is far** from pretraining (vocabulary overlap predicts this).
- **Relevance is the mechanism, not data volume:** adapting to an *irrelevant* domain can drop you *below* the baseline.
- **TAPT** (the task's own unlabeled text) is tiny and $\sim 60\times$ cheaper, yet strong; **DAPT + TAPT** is best.
- Augmenting TAPT with curated / retrieved in-domain text helps more; TAPT itself is task-specific and does not transfer across tasks.

The takeaway: a strong pretrained LM is not the finish line. A little continued in-domain pretraining is *cheap* and *reliably* pays off – so don't stop pretraining.